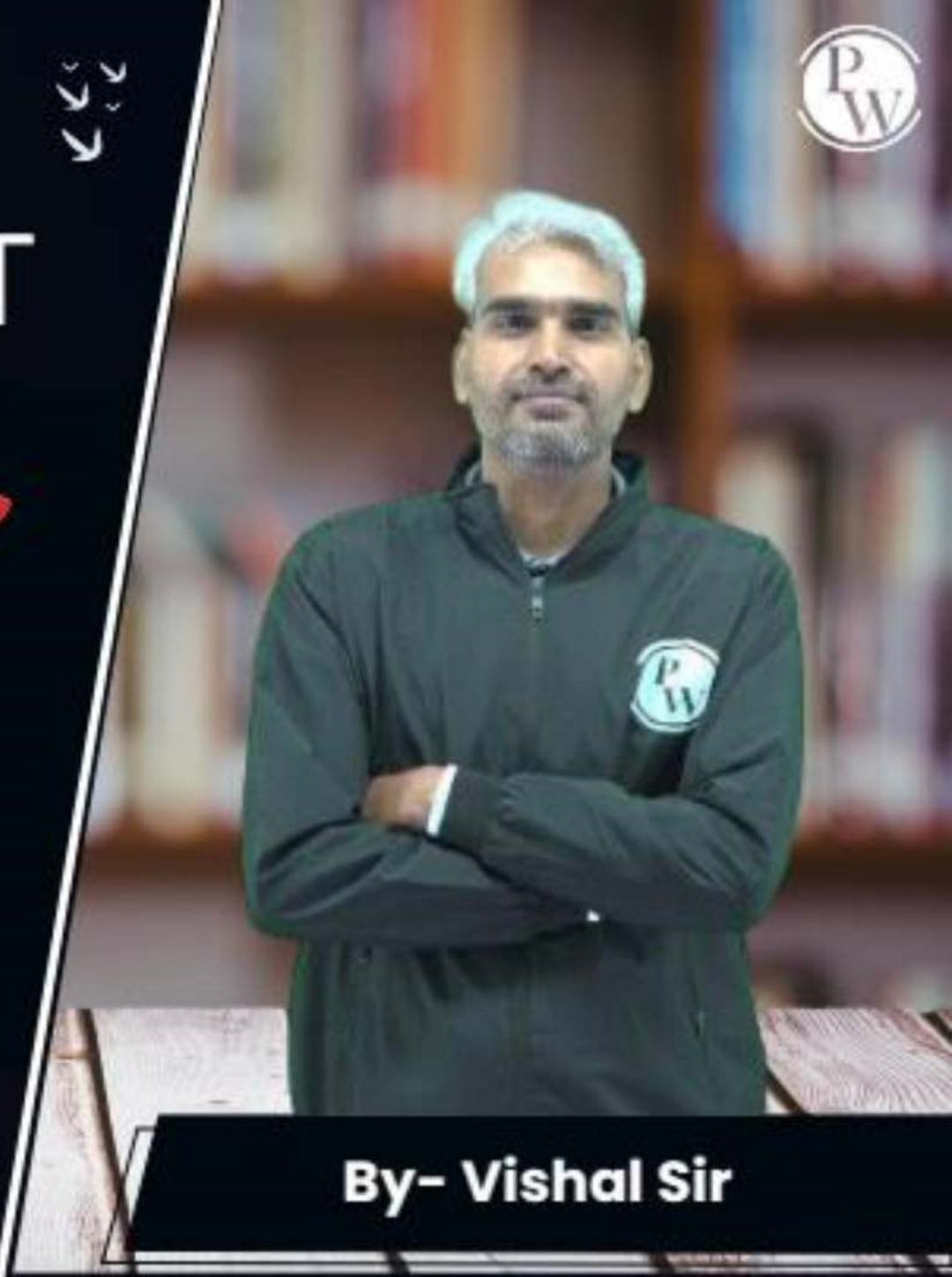
Computer Science & IT



Discrete Mathematics

Set Theory & Algebra

Lecture No. 22



By- Vishal Sir

Recap of Previous Lecture



Topic — Basic concepts associated with group

Topic — Finite group

Topic — Addition modulo ‘m’

Topic — Multiplication modulo ‘m’

Slide

Topics to be Covered



Topic — Groups w.r.t. Addition modulo ‘m’

Topic — Groups w.r.t. Multiplication modulo ‘m’

Topic — Order of an element in a group

Topic — Sub-group

Slide

HoWo

#Q. Let $A = \{0, \pm 2, \pm 4, \pm 6, ...\}$

$B = \{0 \pm 1, \pm 3, \pm 5 .....\}$

Which of the following is not a semi- group

A $(A, +)$ — Associative ✓, Closed ✓

B $(A, \bullet)$ — Associative ✓, Closed ✓

C $(B, +)$ — Closed ✗, Associative ✓

D $(B, \bullet)$ — Closed ✓, Associative ✓

H.W.

$(\Sigma^*, \star)$ — Concatenation

#Q. Consider the set Σ^* of all strings over the alphabet $\Sigma = \{0, 1\}$. Σ^* with the concatenation operator for strings

A. Not a semigroup

B. Semi group but not a monoid

C. Monoid but not a group.

D. A group

① Closed ✓ (∴ groupoid)

② Associative ✓ (∴ Semigroup)

③ $\epsilon \in \Sigma^*$ (NULL string) ∴ Identity ✓ (∴ Monoid)

④ $inv(\epsilon) = \epsilon$

inverse does not exist for any other string of set Σ^* Except for 'ϵ'. ∴ Not a group.

HW

#Q. Let A be the set of all non-singular matrices of order n×n over real number and let * be the matrix multiplication operation. Then

i.e.; determinant value is "not Zero"

(A) A is closed under * but ⟨A ,*⟩ is not a semigroup

(B) ⟨A ,*⟩ is a semigroup but not a monoid.

(C) ⟨A ,*⟩ is a monoid but not a group.

(D) ⟨A ,*⟩ is a group but not an abelian group.

① Closed ✓ (∴ groupoid)

② Matrix multiplication is associative ✓ (∴ Semigroup)

③ Identity matrix of order 'n×n' is also a non-singular matrix ∴ belongs to set A ✓ (∴ Monoid)

④ Inverse of a non-singular matrix of order: n×n, is also a non-singular matrix of order 'n×n'

∴ Inverse for every matrix of set A is present in set A itself (∴ group)

Matrix multiplication is not commutative

H.W.

#Q. Let S be any finite set, and F(s) is defined as set of all function on set S. Then F(s) with respect to function composition operation (ie., o) is.

∴ if $f_1 \in F(S)$ then 'f_1' is a fn from 'S' to 'S'

A. Not a semigroup

B. Semi group but not a monoid

C. Monoid but not a group.

D. A group

① let $f_1, f_2 \in F(S)$

$\therefore f_1 : S \to S$ & $f_2 : S \to S$

Hence; $f_1 \circ f_2 : S \to S \therefore f_1 \circ f_2 \in F(S)$

& $f_2 \circ f_1 : S \to S \therefore f_2 \circ f_1 \in F(S)$

∴ Closed (Groupoid)

② Function Composition is associative ∴ (Semigroup)

③ Identity function on set S, i.e.; $I_S \in F(S)$

s.t. if $f \in F(S)$, then $f \circ I_S = f$ & $I_S \circ f = f$

∴ Identity exist Hence (Monoid)

④ All functions on set S need not be bijective

∴ Inverse will not exist for function that are not bijective (∴ Not a group)

H.W.

#Q. Let Z is the set of all integers. The binary operation * is defined as a*b = max (a, b) then the structure (Z,*) is

A. Not a semigroup

B. Semi group but not a monoid

C. Monoid but not a group.

D. A group

$a*b = \max(a,b)$

① if $a,b \in Z$, then $\max(a,b) \in Z$ ∴ Closed ∴ (groupoid)

② $(a*b)*c$ $\quad a*(b*c)$

$\max(\max(a,b), c)$ $\quad = \quad \max(a, \max(b,c))$

o/p: Max of a, b & c $\quad$ o/p: Max of a, b & c

③ Identity element will be the smallest integer, and it does not exist ∴ Identity does not exist

∴ Associative (Hence, semigroup)

∴ Not a monoid.

H.W.

#Q. Let Q^+ be the set of all positive rational numbers. The binary operation $*$ is defined as $a * b = \frac{ab}{3} \forall a, b, \in Q^+$ If $(Q^+, *)$ is a group then find

(i) identity element of the group $\longrightarrow$?

(ii) inverse of any element a, $\forall a \in$ Group

let 'e' is identity,

We know $\forall a \in Q^+$, $a * e = a$

$\Rightarrow \frac{a \cdot e}{3} = a$

$\boxed{e = 3}$

$a * (a^{-1}) = e$

$\Rightarrow \frac{a \cdot (a^{-1})}{3} = 3 \Rightarrow a \cdot a^{-1} = 9$

$\therefore \boxed{a^{-1} = 9/a}$

H.W.

#Q. Which of the following statement is/are not true.

$x \in Z$

inverse $(2xk) = -2xk$

Closed ✓ | '+' is associative | Identity w.r.t. '+' is '0' ∉ 0 ∈ Set ∴ Identity exist

A {0, ± 2k, ± 4k,, ±6k, ...} is a group with respect to addition where k is any fixed positive integer (True)

B {x | x is real number and $0 < x \leq 1$} is a group with respect to multiplication (False)

Closed ✓ | Multiplication is associative ✓ | Identity w.r.t. Multiplication = 1 ∈ Set ∴ Identity exist ✓

• Inverse does not exist for any other element, except for '1' ∴ Not a group

C $\{2^n \mid n \text{ is an integer}\}$ is a group with respect to multiplication (True)

Set = $\{\ldots, 2^{-3}, 2^{-2}, 2^{-1}, 2^0, 2^1, 2^2, 2^3, \ldots\}$

① If $n_1, n_2 \in Z$ then $2^{n_1}, 2^{n_2} \in$ Set

& $2^{n_1} \cdot 2^{n_2} = 2^{n_1+n_2}$ if $n_1, n_2 \in Z$ then $(n_1+n_2) \in Z$ ∴ $2^{n_1+n_2} \in$ Set Hence "Closed"

② Multiplication is associative

③ $2^0 = 1 \in$ Set ∴ Identity w.r.t. Multiplication exist in the set

④ let $n_1 \in Z$ inv $(2^{n_1}) = 2^{-n_1} \in$ Set



Topic : Addition modulo 'm' $\oplus_m$

- Let 'm' is a fixed positive integer. (it gives remainder)
- For any two non-negative integers a & b

$$a \oplus_m b = \begin{cases} a+b & \text{if } (a+b) < m \\ r & \text{otherwise} \end{cases}$$

$\Leftrightarrow (a+b) \% m$

r is the remainder obtained on dividing $(a+b)$ by m

eg: $2 \oplus_6 3 = 5$

$4 \oplus_6 5 = 3$



Topic : Multiplication modulo 'm' $\otimes_m$

- Let m is a fixed Positive integer (it gives remainder)
- For any two non-negative integer a & b

$$a \otimes_m b = \begin{cases} (a \cdot b) & \text{if } (a \cdot b) < m \\ \text{or} & \text{Otherwise} \\ (\text{remainder}) & \end{cases}$$

$$\Downarrow$$

$$(a \cdot b) \,\%\, m$$

Eg: $2 \otimes_7 3 = 6$

$4 \otimes_7 5 = 6$



Topic : NOTE

① The set $\{0, 1, 2, \ldots\ldots, (n-1)\}$ form a finite group of order $= n$ w.r.t. binary opn $\oplus_n$

eg: $\{0, 1, 2, 3, 4\}$ is a finite group of order $= 5$ w.r.t $\oplus_5$

✓ ① Closed

✓ ② $\oplus_n$ is associative for any integer 'n'

③ Identity w.r.t. $\oplus_n$ is always '0'

④ $inv(0) = 0$ $\quad inv(2) = 3$

$inv(1) = 4$ $\quad inv(3) = 2$

$inv(4) = 1$

$inv(0) = 0$ {because '0' is identity}

for any other $x \in$ Set $\quad inv(x) = n - x$ w.r.t. $\oplus_n$

Note:-

set of element 'x' / such that / $x \in N$ and $x < N$ and — i.e., x & n are Co-prime or 'relatively prime'

let $A = \{ x \mid x \in N \text{ and } x < n \text{ and } GCD(x, n) = 1 \}$

↳ Here, A is a set of all natural numbers that are less than 'n' and Co-prime to 'n'

Eg: let $n = 15$ then $A = \{1, 2, 4, 7, 8, 11, 13, 14\}$

↳ A is a set of all natural numbers that are less than '15' & Coprime to '15'



Topic : NOTE

- The set of all natural numbers that are less than 'n' and coprime to 'n' form a finite group w.r.t. $\otimes_n$

{multiplication modulo 'n'}

eg: let $n = 15$, then

$\{1, 2, 4, 7, 8, 11, 13, 14\}$ is group of order = 8 w.r.t. binary op^h $\otimes_{15}$

① Closed ✓

② $\otimes_n$ is associative for any positive integer 'n'.

③ w.r.t. $\otimes_n$, identity is always = 1

④ Inverse:

$inv(1) = 1 \Rightarrow 1 \otimes_{15} 1 = 1$ (identity)

$inv(2) = 8 \Rightarrow 2 \otimes_{15} 8 = 1$ (identity)

$inv(4) = 4 \Rightarrow 4 \otimes_{15} 4 = 1$ (identity)

$inv(7) = 13 \Rightarrow 7 \otimes_{15} 13 = 1$ (identity)

$inv(8) = 2$

$inv(11) = 11 \Rightarrow 11 \otimes_{15} 11 = 1$ (identity)

$inv(13) = 7$

$inv(14) = 14 \Rightarrow 14 \otimes_{15} 14 = 1$ (identity)

eg. Let $n = 8$, then

$\{1, 3, 5, 7\}$ is a finite group of order $= 4$ w.r.t. $\otimes_8$

1. Closed ✓
2. $\otimes_4$ is associative ✓
3. Identity $= 1 \in$ Set ✓
4. Inverse

$inv(1) = 1$

$inv(3) = 3$

$inv(5) = 5$

$inv(7) = 7$

In this group, Every element happens to be inverse of itself.

eg: ① $\{1, 3, 5, 7\}$ is a group of order = 4 w.r.t. $\otimes_8$

Note: * $\{1, 3\}$ is a finite group of order: 2 w.r.t. $\otimes_8$

* $\{1, 5\}$ is a finite group of order: 2 w.r.t. $\otimes_8$

* $\{1, 7\}$ is a finite group of order: 2 w.r.t. $\otimes_8$

← some subsets of original set may also be the group w.r.t. same binary opⁿ

→ But all subsets need not be the group w.r.t. same binary operation.

eg: $\{3, 5\}$ is not a group w.r.t. $\otimes_8$

eg: $\{1, 3, 5\}$ is not a group w.r.t. $\otimes_8$

eg: We know $\{0, 1, 2, 3\}$ is a group w.r.t. $\oplus_4$

Here, • $\{0\}$ is also a group w.r.t. $\oplus_4$

• $\{0, 2\}$ is also a group w.r.t. $\oplus_4$

• $\{0, 1\}$ is not closed w.r.t. $\oplus_4$ ∴ not a group

$1 \oplus_4 1 = 2 \in \{0, 1\}$

∴ Not closed Hence not a group.

Note:- We know,

The set of all natural numbers that are less than n and Co-prime to n form a group w.r.t. binary op^n $\otimes_n$

∴ if 'p' is a prime number then the set $\{1, 2, 3, \ldots\ldots, (p-1)\}$ will form a finite group of order $= (p-1)$ w.r.t. binary op^n $\otimes_p$

eg: The set $\{1, 2, 3, 4, 5, 6\}$ is a group of order $= 6$ w.r.t. op^n $\otimes_7$

Q: Which of the following is/are not true?

(a) $\{1, 2, 3, 4, 5\}$ is a group w.r.t. $\oplus_6$ (Not true)

identity = $0 \notin$ Set $\therefore$ can not be group w.r.t. $\oplus_6$

(b) $\{1, 2, 3, 4, 5\}$ is a group w.r.t. $\otimes_6$ (Not true)

$2 \otimes_6 3 = 0 \notin$ Set $\therefore$ Not Closed, Hence not a group

(c) $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\oplus_5$ (True)

(True)

(d) $\{0, 1, 2, 3, 4\}$ is a group w.r.t. $\otimes_5$ (Not True)

$0 \in$ Set & inv(0) : Does not exist $\therefore$ Not a group



2 mins Summary

Topic Groups w.r.t. Addition modulo ‘m’

Topic Groups w.r.t. Multiplication modulo ‘m’

Topic Order of an element in a group

Topic Sub-group

THANK - YOU